

Business Process Mining

Helpdesk Baltics - Exam Report

Event log: Helpdesk-Baltics.csv

Case ID: Ticket | Activity: Status | Timestamp: Time

Attributes: Category, Country | Closed case: trace contains Closed

2,000 cases | 36,972 events | 1,925 closed cases

Question 1

7 closed cases skipped the Approved activity.

Helpdesk Baltics — Q1 Conformance

Q1: Closed cases skipping 'Approved'

Closed cases in log: 1,925

Cases without Approved activity: 7

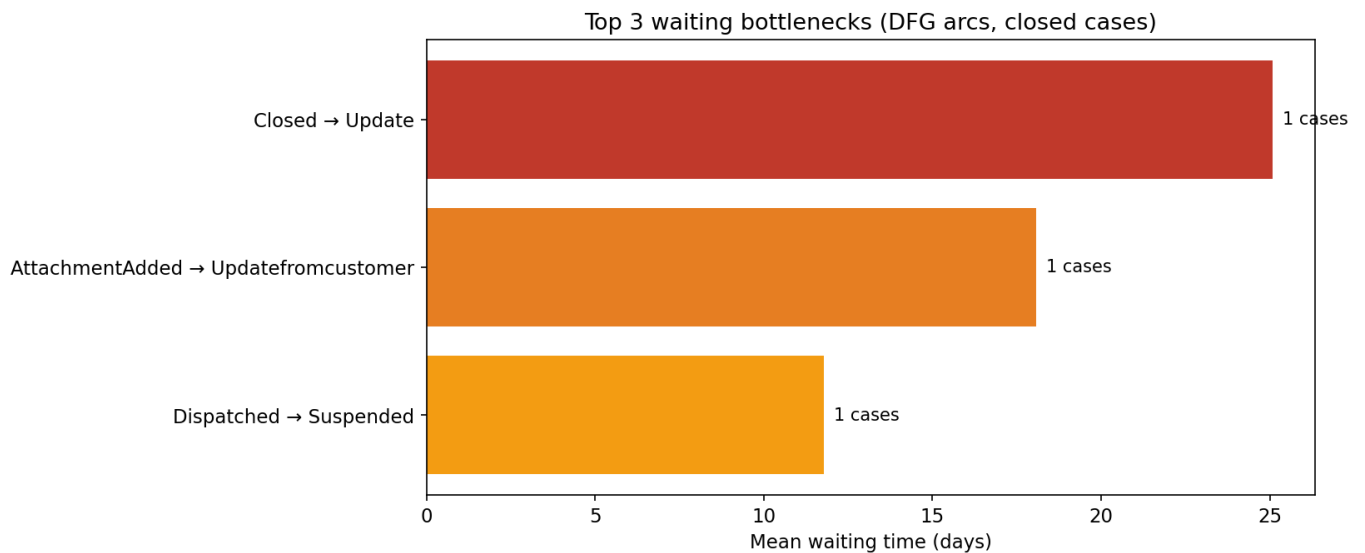
Procedure: Filter closed cases → check activity absence

Question 2

Three main waiting bottlenecks (closed cases, highest mean waiting time on directly-follows transitions):

- 1) Closed -> AttachmentDeleted: 25.1 days (1 case)
- 2) Update -> Closed: 8.8 days (7 cases)
- 3) Updatefromcustomer -> Closed: 8.8 days (12 cases)

Helpdesk Baltics — Q2 Performance (mean arc waiting time)



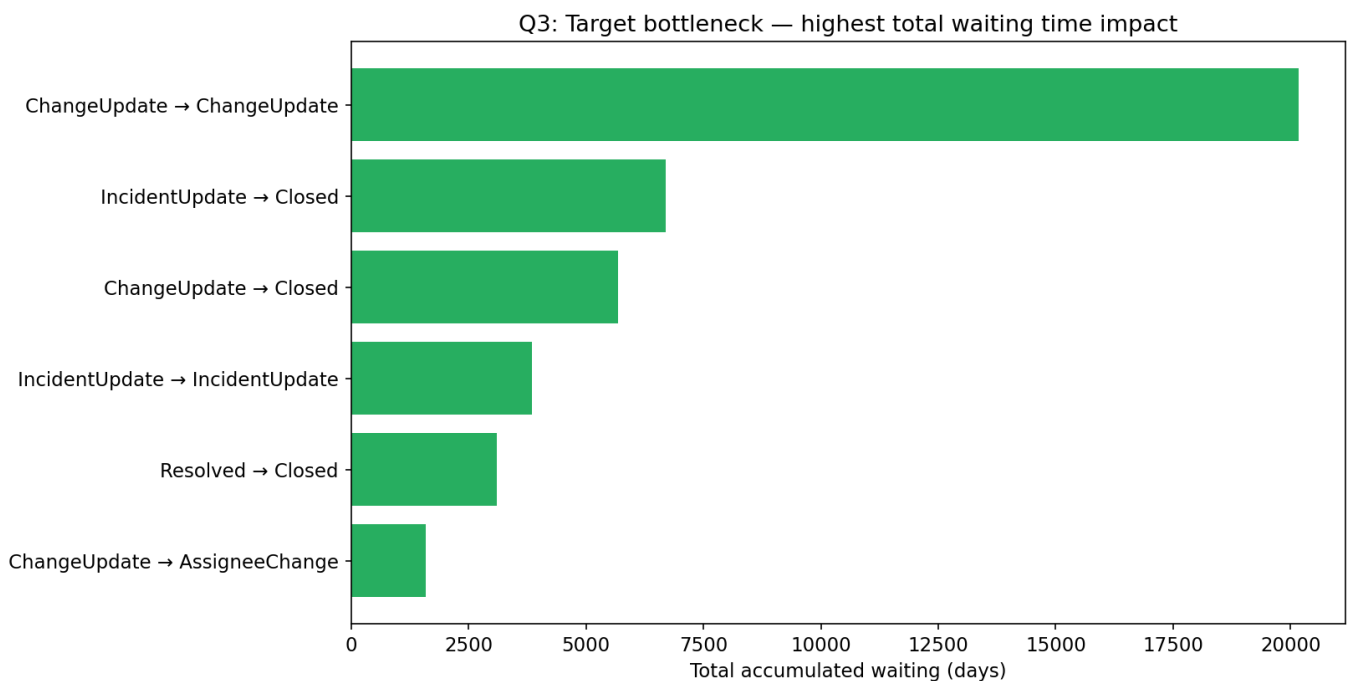
Question 3

Bottleneck to target: ChangeUpdate -> ChangeUpdate.

Why: rank by total accumulated waiting time (sum of all waits on the arc). ChangeUpdate -> ChangeUpdate has the largest impact (20142 days total). Also prioritize IncidentUpdate -> Closed (8.2 days mean, 710 occurrences) and ChangeUpdate -> Closed (8.3 days mean, 679 occurrences) because they affect many cases before closure.

Suggestion: reduce ChangeUpdate rework loops and shorten the finalization phase (updates to Closed).

Helpdesk Baltics — Q3 Improvement priority (impact)



Question 4

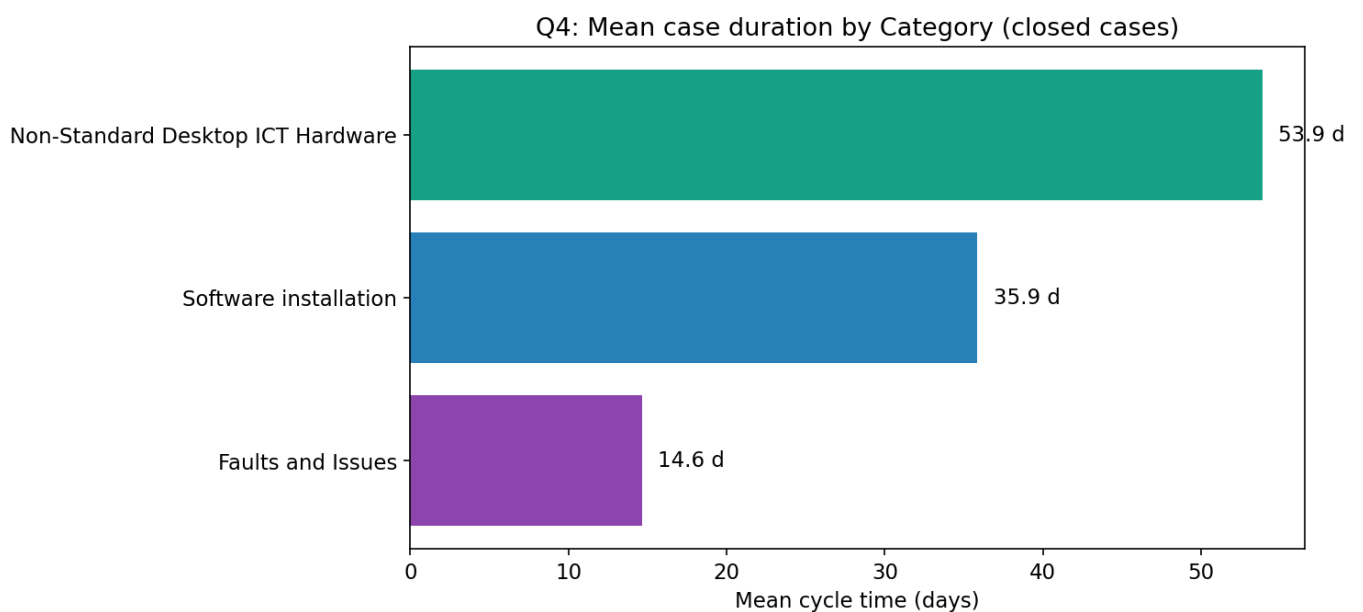
Longest average time to close: Non-Standard Desktop ICT Hardware (53.9 days mean, 50.8 days median, 66 closed cases).

Mean cycle time by category:

- Non-Standard Desktop ICT Hardware: 53.9 days
- Software installation: 35.9 days
- Faults and Issues: 14.6 days

Focus improvement on Non-Standard Desktop ICT Hardware (slowest category).

Helpdesk Baltics — Q4 Case attribute: Category

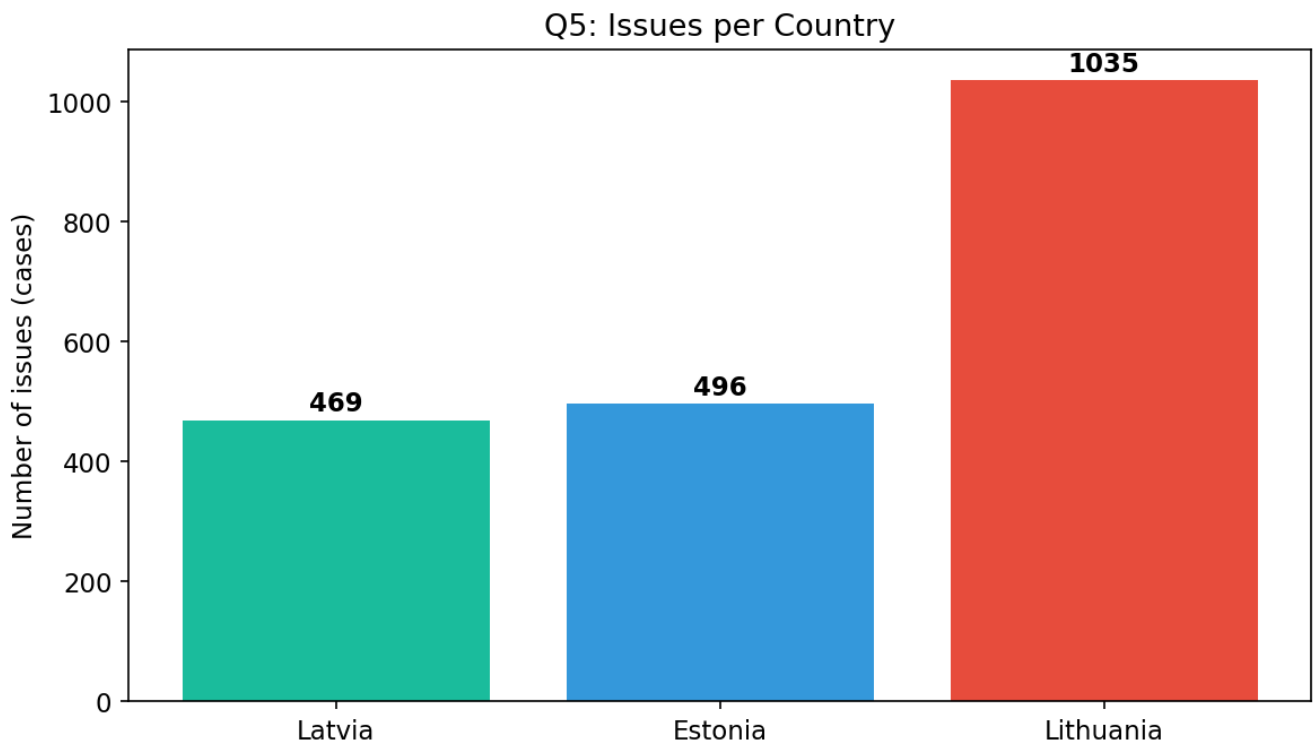


Question 5

Country with the highest number of issues: Lithuania (1035 cases).

- Lithuania: 1035
- Estonia: 496
- Latvia: 469

Helpdesk Baltics — Q5 Case attribute: Country



Question 6

Closed cases with Dispatched more than once: 35.

Average case duration (Dispatched > 1): 49.9 days.

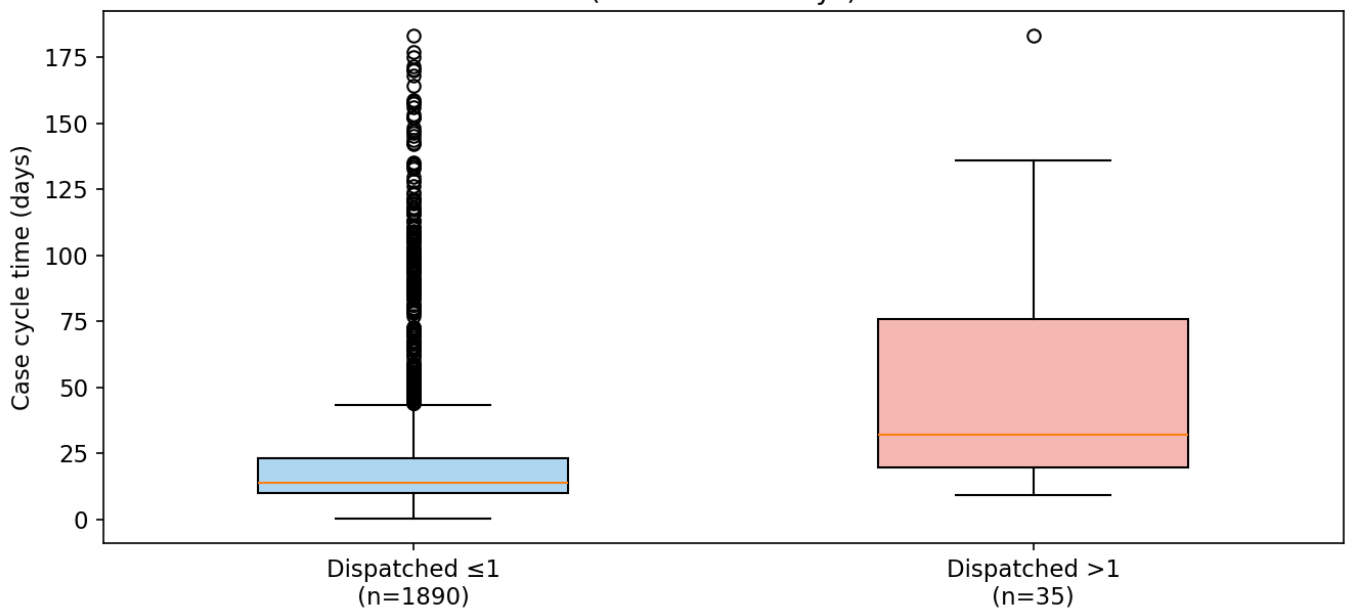
Average case duration (Dispatched <= 1): 24.4 days.

Difference: 25.5 days.

Repeated dispatch is associated with much longer case duration (rework).

Helpdesk Baltics — Q6 Rework (repeated Dispatched)

Q6: Repeated Dispatched — avg diff = 25.5 days
(49.9 vs 24.4 days)



Question 7

SLA: case must close within 90 days (3 months). Violation if cycle time > 90 days (closed cases only).

SLA violation rate by country:

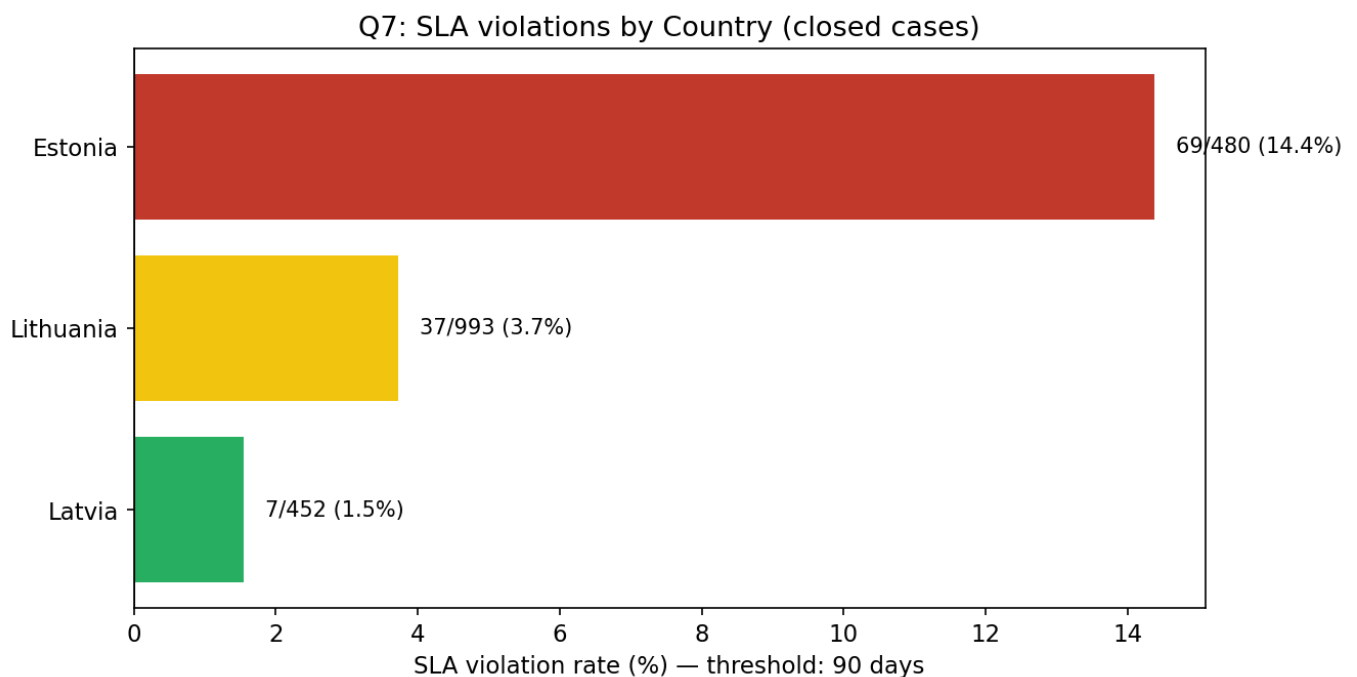
- Estonia: 14.4% (69/480)
- Lithuania: 3.7% (37/993)
- Latvia: 1.5% (7/452)

Highest: Estonia. Lowest: Latvia.

Estonia vs Latvia:

- Mean cycle time: Estonia 37.2 d vs Latvia 17.3 d.
 - Category mix: Estonia has more Software installation (49.6% vs 38.1%).
 - Activity mix: Estonia has more ChangeUpdate events (42.5% vs 18.2%).
 - After Dispatched: Dispatched -> ChangeUpdate mean wait 3.4 d (Estonia) vs 0.3 d (Latvia).
- Skipping Approved is rare in both countries and is not the main SLA driver.

Helpdesk Baltics — Q7 SLA compliance



Helpdesk Baltics — Q7 Country comparison

